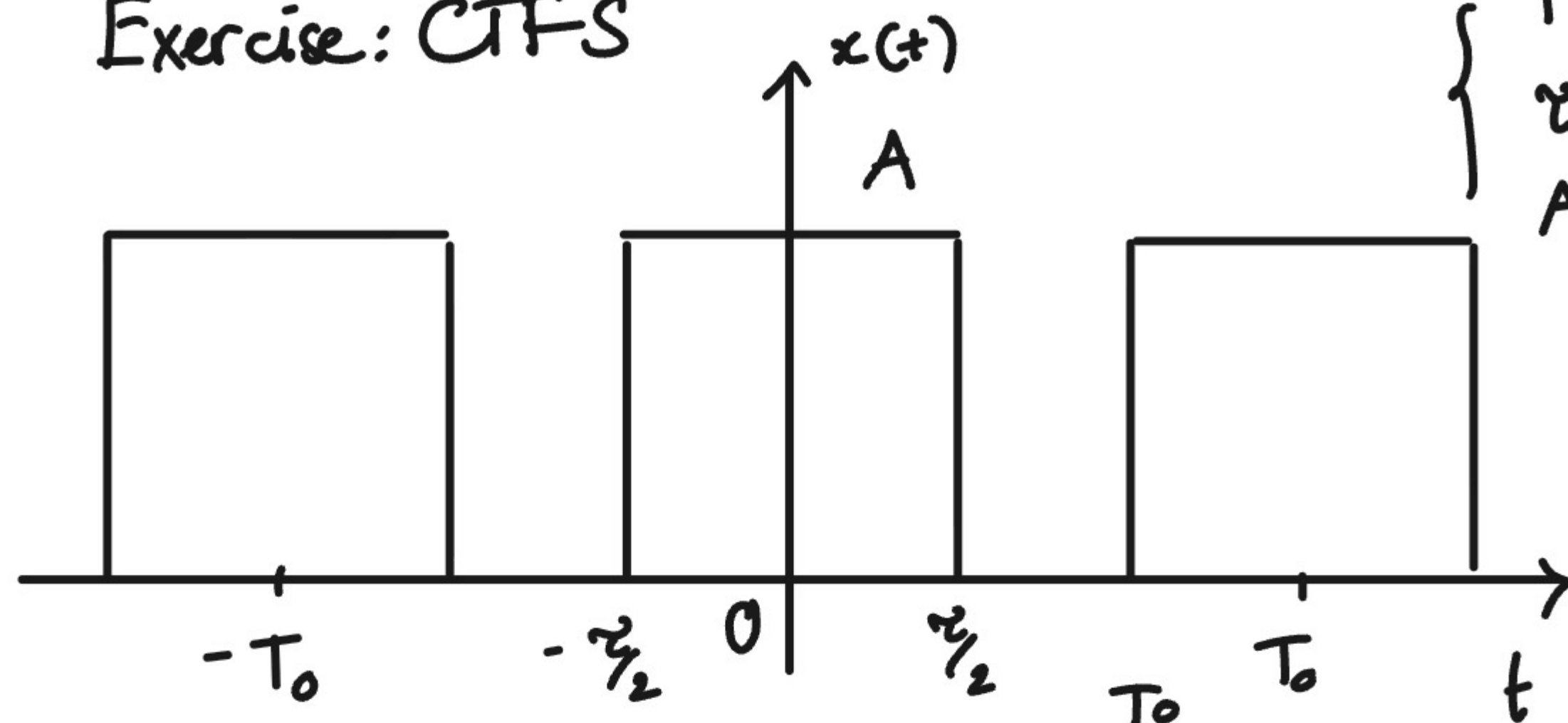


Exercise: CTFs



$$\begin{cases} T_0 = 5s \\ \tau = 1s \\ A = 1 \end{cases} \Rightarrow F_0 = \frac{1}{T_0} = 0,2 \text{ Hz}$$

Turn phas?

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^{-j\omega_0 k t} dt = \frac{1}{T_0} \int_{-\frac{T_0}{2}}^{\frac{T_0}{2}} x(t) \cdot e^{-jk\omega_0 t} dt = \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} x(t) \cdot e^{-jk2\pi F_0 t} dt$$

$$= \frac{1}{T_0} \int_{-\tau/2}^{\tau/2} A e^{-jk2\pi F_0 t} dt = \frac{A}{T_0} \cdot \frac{1}{-jk2\pi F_0} \cdot e^{-jk2\pi F_0 t} \Big|_{-\tau/2}^{\tau/2}$$

$$e^{j\phi} - e^{-j\phi} = 2j \sin \phi$$

$$= \frac{A}{T_0} \cdot \frac{1}{-jk \cdot 2\pi F_0} \cdot \left(e^{-jk2\pi F_0 \cdot \tau/2} - e^{+jk2\pi F_0 \cdot \tau/2} \right)$$

$$f(x) = \text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$$

$$= \frac{A}{T_0} \cdot \frac{1}{-jk2\pi F_0} \cdot 2j \sin(-k2\pi F_0 \tau/2) = \frac{A}{T_0} \cdot \frac{\sin(-k\pi F_0 \tau)}{k\pi F_0} = \frac{A}{T_0} \cdot \frac{\sin(\pi(-kF_0 \tau))}{(-kF_0 \tau) \cdot \pi} \cdot -\tau$$

$$= \frac{A}{T_0} \text{sinc}(-kF_0 \tau) \cdot -\tau$$

thay $A = 1, T_0 = 5, \tau = 1$, ta đươc:

$$c_k = -\frac{1}{5} \cdot \text{sinc}(-k \cdot 0,2 \cdot 1) = \frac{1}{5} \text{sinc}(0,2k)$$